BAMAT2O7 — PROBABILITY AND STATISTICS

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ASSESSMENT-4

Question:

Create a data frame containing the given observations,

20.1, 19.8, 20.3, 20.0, 19.7, 20.2, 20.1, 19.9, 20.4, 19.8, 20.0, 20.2, 19.6, 20.3, 19.9, 20.1, 20.0, 19.7, 20.2, 19.8

Compute the following measures of dispersion:

- Range
- Variance
- Standard deviation
- Compute the measure of skewness.

Solution:

Console | Terminal | Background Jobs

R 4.6.1 · ~/

```
> data <- data.frame(
+     observations = c(
+         20.1, 19.8, 20.3, 20.0, 19.7,
+         20.2, 20.1, 19.9, 20.4, 19.8,
+         20.0, 20.2, 19.6, 20.3, 19.9,
+         20.1, 20.0, 19.7, 20.2, 19.8
+     )
+ )
>
> data
   observations
1          20.1
2          19.8
3          20.3
4          20.0
5          19.7
6          20.2
7          20.1
8          19.9
9          20.4
10         19.8
11         20.0
12         20.2
13         19.6
14         20.3
15         19.9
16         20.1
17         20.0
18         19.7
19         20.2
20         19.8
```

```
> range(data$observations)
[1] 19.6 20.4
> diff(range(data$observations))
[1] 0.8
> var(data$observations)
[1] 0.05102632
> sd(data$observations)
[1] 0.2258901
> install.packages("e1071")
```

```
WARNING: Rtools is required to build R packages but is not currently installed. Please download an
d install the appropriate version of Rtools before proceeding:

https://cran.rstudio.com/bin/windows/Rtools/
```

```
Installing package into 'C:/Users/DELL/AppData/Local/R/win-library/4.6'
(as 'lib' is unspecified)
also installing the dependency 'proxy'
```

```
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.6/proxy_0.4-29.zip'
trying URL 'https://cran.rstudio.com/bin/windows/contrib/4.6/e1071_1.7-17.zip'
package 'proxy' successfully unpacked and MD5 sums checked
package 'e1071' successfully unpacked and MD5 sums checked

The downloaded binary packages are in
        C:\Users\DELL\AppData\Local\Temp\Rtmp8kbPQ3\downloaded_packages
> library(e1071)
> skewness(data$observations)
[1] -0.05875674
>
```

Code for reference-->

```
data <- data.frame(
  observations = c(
    20.1, 19.8, 20.3, 20.0, 19.7,
    20.2, 20.1, 19.9, 20.4, 19.8,
    20.0, 20.2, 19.6, 20.3, 19.9,
    20.1, 20.0, 19.7, 20.2, 19.8
  )
)


range(data$observations)
diff(range(data$observations))
var(data$observations)
sd(data$observations)


install.packages("e1071")
library(e1071)
skewness(data$observations)
```